House Price Developments in Austria and Germany: Evidence from the Latest HFCS Wave

Course

Current Topics in Macroeconomic Policy (MA)

February 13, 2025

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Acknowledgment:

This paper uses data from the Eurosystem Household Finance and Consumption Survey.

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1. Introduction

The housing market plays a critical role in macroeconomic stability. A good example of the extent of its importance can be found in the events leading up to the Financial Crisis of 2007-2008. In fact, and as pointed out by Girouard and Blöndal (2001), house price developments are associated in a substantial way with the business cycle. Relevant macroeconomic variables and households' decisions regarding consumption, investments and savings can be heavily influenced by fluctuations in house prices. As a matter of fact, for a significant share of households their main residences represent the most valuable non-financial asset they own. Furthermore, the value of these assets is systematically taken by credit institutions as a reliable collateral determining their access to credit. All things considered, understanding house price developments provides valuable insights into broader economic trends and serves as a tool to policymakers to conduct macroprudential policies.

This seminar paper examines house price developments in Austria and Germany with the help of the latest available data from the Household Finance and Consumption Survey (HFCS) 2021 wave. The HFCS is a survey which is coordinately conducted by the national banks and relevant statistical institutes of the country members of the European Union (EU) under the supervision and guidance of the European Central Bank (ECB). The information extracted from the questionnaire yields a rich source of microdata at the household level regarding their socioeconomic characteristics, their balance sheet -including the properties their own and their elicited prices- and, overall, their finance and consumption habits. This is a source of information which in principle allows for cross-country, harmonized comparisons.

In order to conduct our study, the statistical analysis has been split into three main sections. Firstly, we calculate the change in real estate values in each country by comparing initial purchase prices to current market values. Secondly, we identify which households have experienced significant increases in their real estate assets values in terms of some of their socioeconomic characteristics. Lastly, we analyze whether these households have adjusted their consumption and savings behavior in response to the rise in their real estate wealth. Across all sections, we compare both countries to identify commonalities and differences. A detailed account of the implemented methodology will be provided in further sections of the paper.

Our contribution to the literature builds upon the fact that our analysis uses high-quality survey data to study changes in property prices that were acquired as far back as 1940, which ultimately allows us to give valuable insights into a wide variety of household profiles in both countries.

Our main findings suggest through the lens of the HFCS that around 43,3% of households in Austria and 37,2% in Germany have suffered a 100% increase in value of their main residences. Also, households with higher net worth and net liquid assets in which their reference person belongs to the age brackets of 66 to 75 and 76 to 85 years old are the ones who tend to have

experienced the most changes in value in both countries. Our analysis does not provide any evidence suggesting that the households who experienced in their real estate wealth have adjusted their consumption and/or savings behavior.

The rest of our seminar paper will be organized as follows: section (2) presents the main theoretical foundations of our analysis with the help of a review of the most relevant literature; section (3) studies the changes in real estate values with a focus on main residences; section (4) identifies which households have suffered the most significant real estate value changes; section (5) analyzes the consumption and saving behavior of the households that have experienced the most change in value of their main residences; in section (6) we discuss our results and their policy implications and section (7) concludes. It must be noted that for each section (3), (4) and (5) we first motivate the analysis, then we give an account of the used variables and methodology and finally we present the results.

2. Theoretical Background

As previously mentioned, house price developments play a crucial role in shaping household wealth and overall financial stability. The importance of house price developments has been broadly studied in the literature. As pointed out by Girouard and Blöndal (2001), they are associated in a substantial way with the business cycle and, as they find, house prices in OECD countries significantly influence private consumption by way of wealth effects or by simply relaxing households' liquidity constraints. Also, Albacete and Lindner (2017) examined the effects of macroprudential policies on the housing sector, showing how regulations can influence mortgage borrowing and market stability. They find a positive but weak relationship between house prices and credit supply in Austria. Relatedly, Jarocinski and Smets (2008) find housing demand shocks to affect residential investments and house prices in the US and give insights into the role of monetary policy on residential investments. Overall, these findings highlight the relevance of house price developments in terms of policy implications and in ongoing aggregate demand trends.

A wide variety of methodologies have been proposed to evaluate house price developments. Philiponnet and Turrini (2017) discuss the convenience of using time series of price-to-income and price-to-rent ratios to assess how balanced house price developments are in terms of their macroeconomic risk implications. Albacete et al. (2016) construct average transaction and appreciation house price indices using HFCS data and study the financial stability implications of house price developments in Austria. They focus on the price developments of household main residences but also include the set of most important other properties in their analysis, which are quantitatively less relevant, a reason why we won't take them into account. Although the HFCS reports non-transactional house property values, which for the most part don't accurately represent the actual market value of a property because no property transaction has taken place

at that price, they conclude that the HFCS house price indices they construct are valid in that they track within acceptable margins other relevant house price indices in Austria, such as the TU/OeNB house price index. They also find that most households experienced a yearly change in value in their main residences of 0% to 5%.

One major question in housing economics is how property value growth affects household decision-making. Kindermann et al. (2021) investigated how people learn about housing costs and adjust their financial choices accordingly. Their research suggests that when property prices rise significantly, households may shift their consumption habits: some might use their increased equity to spend more, while others might adjust their savings in response to future expectations. These authors also use survey data in Germany to document how house values have changed in the period 2005-2020 and find an overall increasing price trend, with the region of Southern Germany showing particularly high growth.

Schneider and Wagner (2015) provide some stylized facts regarding house price developments in Austria and Germany. The authors observe acute house price increases in Austria since 2005, with nominal prices rising by 44% from early 2007 until late 2014. In Germany they observe a 1,3% annual increase in nominal house prices from 1980 until 2013. They attribute the rise in prices to several factors, high demand for safe assets and low returns on financial investments among others. In their paper and in contrast with what we do, they also depict the evolution of real house prices, as they only consider property values from 1970 onwards. A similar approach that takes real prices into account is also proposed by Philiponnet and Turrini (2017). The property price values we work with in the HFCS go from 1940 to 2021, making the study of real price developments challenging from a methodological point of view due to the lack of robust and complete HICP indices for Austria and Germany, mainly due to historical reasons.

Another string of literature tries to identify the mechanisms driving house price changes. They focus on whether these changes are justified by so-called "house price fundamentals" i.e. structural economic factors which ultimately determine the price of a house in the long-run, such as disposable income or interest rates, or rather by market trends with underlying speculative motives that typically lead to "bubbles". Schneider (2013) in a study for Vienna and Austria concludes that some of the variation in house prices in Vienna could be due to such speculative motives. Belke and Keil (2017) however, in a regional study for Germany find that house price developments have been driven mainly by fundamental motives.

Lastly, we will interpret the changes in house property values at the household level following the life-cycle asset accumulation profile proposed by Kaplan and Violante (2014).

Building on these insights, we aim not only to examine how real estate values have changed but also to look at whether households experiencing major price increases have altered their spending or savings behavior. Austria and Germany serve as an interesting comparison, given

their differing financial and regulatory environments. By analyzing these trends, this research contributes to a broader understanding of housing markets, wealth accumulation, and economic decision-making in European households.

3. Changes in Real Estate Values

The real estate market plays a crucial role in household wealth, and understanding how property values evolve over time helps reveal broader economic trends. In this part of our analysis, we explore how residential property values have changed in Austria and Germany, focusing on both relative and absolute price increases. By comparing current property values with their original purchase prices, we identify patterns of appreciation and assess how widespread significant gains or losses are.

Beyond just tracking price changes, we also consider the distribution of appreciation across different households and acquisition periods. Some properties may have experienced steady, moderate growth, while others saw sharp increases or even declines. By examining these dynamics, we gain a deeper understanding of how property values evolve and what this means for households in Austria and Germany.

3.1 Variables and methodology

For the purpose of calculating the changes in real estate values, we chose 2 main variables: HB0900 and HB0800, which are "Current price of household main residence" and "Property value at the time of its acquisition", respectively.

Firstly, we calculated two key metrics: the "change ratio", which indicates relative price change (current value divided by acquisition value) and the "change absolute", which represents the absolute monetary difference between current and acquisition values. In addition, we introduced the "change category", which groups households into certain bins (such as "Increase $> 100\%$"). In the article that we reference (Albacete et al. (2016)), the authors also analyze the distribution of residential property price changes, focusing on both absolute and relative changes.

After ensuring that "change category" is correctly factored, we checked for and handled missing values: approximately 46.7% of the rows in the data set have missing values (NA) for both HB0800 (acquisition value) and HB0900 (current value). This circumstance can be due to households without real estate properties as not all households in the data set may own such assets, so they will not have values for HB0800 or HB0900. That is why we removed rows with NA to ensure a valid analysis.

Secondly, we calculated descriptive statistics for real estate value changes, such as mean, median, and standard deviation for both "change ratio" and "change absolute", grouped by country. The main result is displayed in Table 1.

Table 1: Summary Statistics for Real Estate Value Changes

	Change Ratio			Change Absolute			Count
Country	**Mean**	**Median**	**SD**	**Mean**	**Median**	**SD**	
AT	7.00	1.82	26.6	179 289	110 000	268 560	4915
DE	4.55	1.88	27.4	232 454	160 000	343 793	12 160

Data: HFCS 2021 wave. Own calculations. SD: standard deviation.

We can clearly observe that the mean change ratio is significantly higher than the median in both countries, particularly in Austria (mean: 7.00 vs. median: 1.82). This suggests the presence of outliers, households whose properties experienced exceptionally large increases in value. The median change ratio indicates that most households in both countries experienced property value increases of 82% (Austria) and 88% (Germany) compared to their acquisition values. The mean absolute price increase is higher in Germany (€232,454) than in Austria (€179,289). Similarly, the median increase in Germany (€160,000) exceeds Austria (€110,000), reflecting higher baseline property values in Germany. Finally, more households were analyzed in Germany (12,160) compared to Austria (4,915).

Thirdly, after this step, we created the "change category" variable within each imputed data set before constructing the survey design object. These categories further allow us to easily compare price changes across HHs. The result is displayed in the Table 2 below.

Table 2: Distribution of Households by Change Category

Change Category and Country		**Weighted Count**	**Proportion**
Decrease > 50%	AT	60 929	0.0141
Decrease 0-50%	AT	164 731	0.0381
Increase 0-50%	AT	1 462 735	**0.338**
Increase 50-100%	AT	761 410	**0.176**
Increase > 100%	AT	1 871 870	**0.433**
Decrease > 50%	DE	411 085	0.0101
Decrease 0-50%	DE	3 122 481	0.0769
Increase 0-50%	DE	13 039 347	**0.321**
Increase 50-100%	DE	8 943 538	**0.220**
Increase > 100%	DE	15 092 500	**0.372**

Data: HFCS 2021 wave. Own calculations.

As is clearly seen from the Table 2, the overall trend suggests that only a small proportion of households in both countries experienced a decline in property value. Specifically, 1.41% of Austrian households and 1.01% of German households saw their property values decrease by more than 50%, while 3.81% of Austrian households and 7.69% of German households experienced a decline of up to 50%. These figures indicate that **downward shifts in property values were relatively rare in both countries**.

The most common category across both Austria and Germany is a modest increase in property values of 0–50%. In Austria, 33.8% of households experienced this level of appreciation, while in Germany, the proportion is 32.1%. This suggests that, for a significant share of households in both countries, property value increases were relatively moderate.

At the same time, the largest category in both countries is Increase ¿ 100%, meaning households whose property values more than doubled. This category accounts for 43.3% of Austrian households and 37.2% of German households, indicating that **a substantial proportion of households in both countries experienced significant capital gains in their real estate assets**.

As a general trend, the distribution of households across the various categories remains broadly similar between Austria and Germany, suggesting comparable real estate market dynamics in both countries. Although some variation exists in the proportion of households experiencing different levels of change, the overall structure of property value shifts follows a similar pattern. For this trend to be more representative, we have added Figure 1 below:

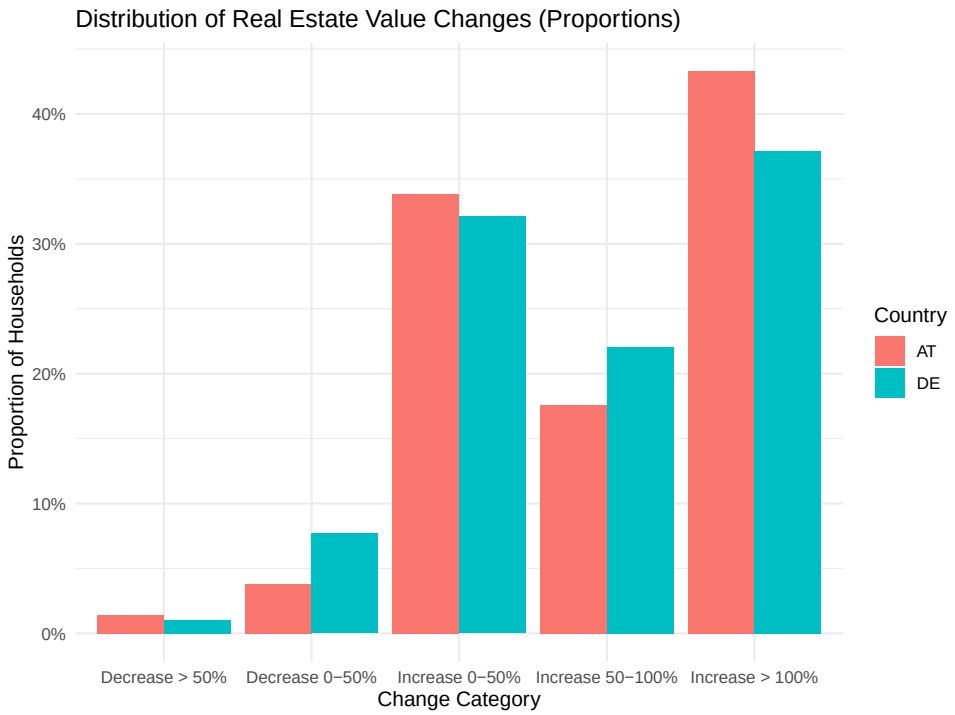

Figure 1
Data: HFCS 2021 wave. Own calculations.

Additionally, we also want to analyze how property value changes vary by year of property acquisition. To do so, we added a year of acquisition category (HB0700 - "Year of property acquisition"), then we grouped the data by these categories, calculated the distribution of property value changes and visualized the results. The distribution of years of acquisition is shown in the Figure 2.

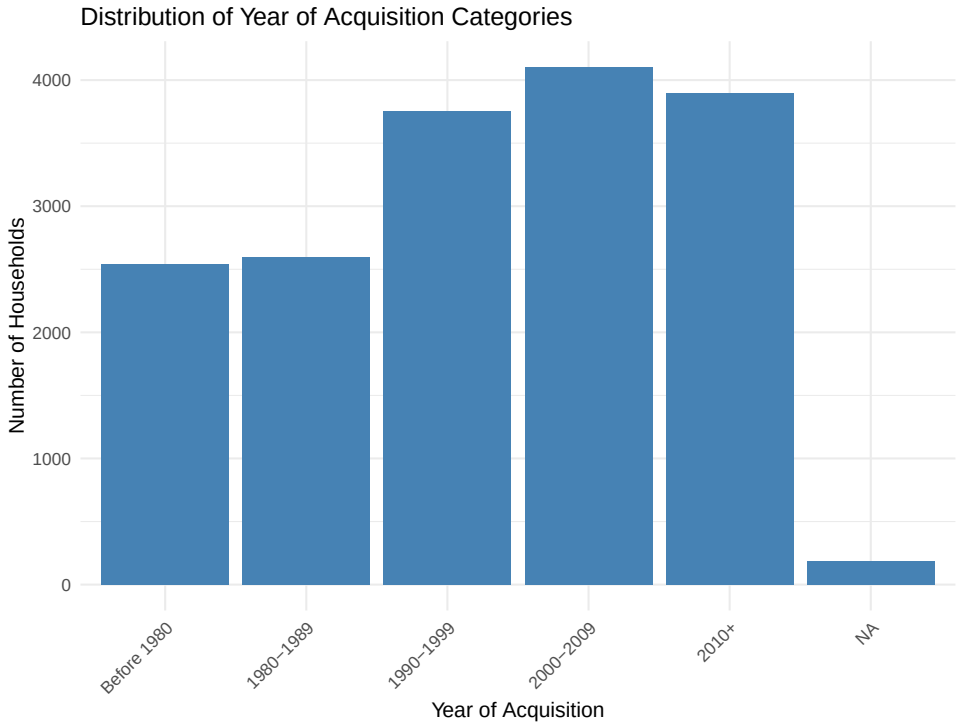

Figure 2
Data: HFCS 2021 wave. Own calculations.

It can be noted that the number of households acquiring properties was relatively low before 1980 but has steadily increased in subsequent decades. A notable peak is observed in recent acquisition periods (2000-2009 and 2010+), indicating that a significant portion of households acquired their properties in recent years. While fluctuations exist between different decades, the overall pattern points to **a growing trend in property acquisitions**.

Finally, we checked the distribution of real estate value changes by year of acquisition for both Austria and Germany. The result is presented in the Figure 3.

As can be seen from this graph, the distribution of real estate value changes by year of acquisition shows that the **"Increase $>$ 100%" category dominates across all acquisition years in both Austria and Germany, particularly for older properties**. However, after 2010, the share of properties experiencing extreme value increases declines, while smaller increases (0-50% and 50-100%) become more common, suggesting a slowdown in appreciation for recent acquisitions. Austria shows greater variability over time, whereas Germany's distribution remains more stable. Notably, property value decreases are minimal in both countries, once again proving **the overall trend of long-term property value appreciation**.

3.2 Results

To conclude, the analysis of real estate value changes in Austria and Germany reveals several key findings. Firstly, **most households in both countries experienced substantial property**

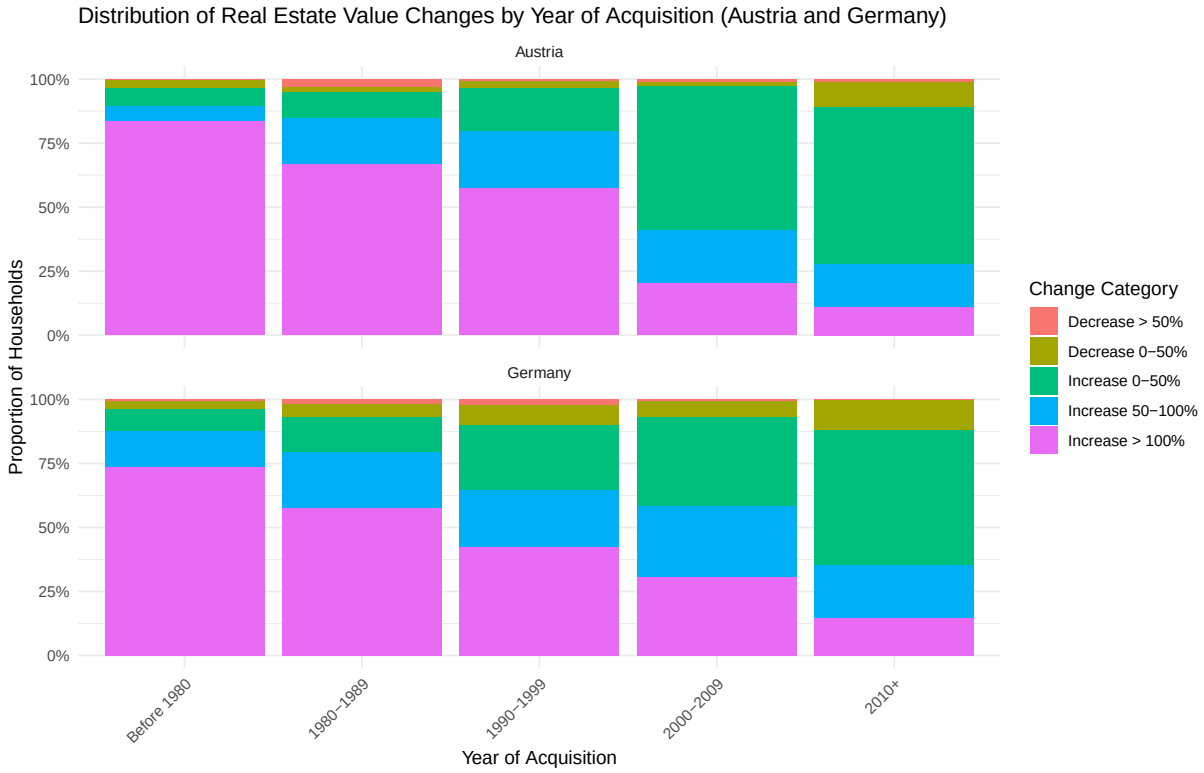

Figure 3
Data: HFCS 2021 wave. Own calculations.

value appreciation, with Austria showing an 82% median increase and Germany an 88% increase relative to acquisition values. However, the mean change ratio is significantly higher than the median in both countries, suggesting the presence of households that experienced extreme increases in property values.

Secondly, absolute price increases were higher in Germany than in Austria, which reflects higher baseline property values in Germany. The distribution of property value changes by category shows that **decreases in property values were relatively rare in both countries**.

The most common category in both countries was a moderate increase of 0-50%, accounting for 33.8% of Austrian households and 32.1% of German households. However, **a substantial proportion of households in both Austria (43.3%) and Germany (37.2%) experienced property value increases exceeding 100%, indicating strong capital gains in real estate**, and this trend remains similar across both countries.

Further analysis by acquisition year shows that **properties acquired before 2010 were more likely to experience extreme value increases (>100%), while those acquired after 2010 saw more moderate growth**. This suggests a slowdown in appreciation for recent acquisitions, particularly in Austria, where variability in property value changes is greater than in Germany. Despite these variations, the overall trend points to long-term property value appreciation in

both countries.

4. Identification of Households with Significant Real Estate Value Changes

After observing how real estate properties have changed in value for the households represented in the survey in Austria and Germany, in this next step we will attempt to identify, according to some of their socioeconomic characteristics and the region where they reside, which households have experienced significant real estate value changes in their main residences.

The process of household identification in this context can provide insightful information to policymakers regarding whether household wealth is becoming bigger between different socioeconomic groups and regions and, accordingly, conduct fiscal policies affecting property taxation, home affordability, inter-generational wealth or specific housing market interventions targeting speculation.

4.1 Variables and methodology

In a preliminary analysis approach, we are interested in studying the role of the net liquid assets and the net worth that a household owns in determining changes in value of their main residences. The economic intuition behind selecting these variables is that we should expect households with more cash-in-hand and/or net worth to own more valuable properties, reflecting a higher purchasing power and wealth. Ultimately, the more valuable a real estate asset is, the more subject it should be to increase in value over time.

According to the HFCS database documentation on core and derived variables, the net liquid assets of a household (coded as "DNNLA") are defined as the "sum of liquid financial assets minus non-collateralised debt" (ECB, 2023, p. 190). The net worth, or "net wealth" as referred to by the ECB, (coded as "DN3001") is defined as the "total household assets excluding public and occupational pension wealth minus total outstanding household's liabilities" (ECB, 2023, p. 189). For both countries, we first compute the main summary statistics of these two variables, followed by an uni-variate regression analysis of each variable against the absolute change in real estate values. A graphical representation of their distribution is also provided. Lastly, we compute Pearson's product-moment correlations to see how linearly correlated they are with the absolute change in value. Even though this statistical tool won't allow us to establish causality in our analysis, we deem it as a appropriate given the preponderant descriptive nature of our study. Nevertheless, we will provide the economic intuition motivating each computation.

Next, we are interested in further delimiting the households' socioeconomic profile. For that, we try to see with which additional variables the absolute change in real estate values could be correlated with. On the one hand and related to the household's reference person, we will mainly focus on two variables that, according to the general economic literature, are important but not exclusive determinants of income, namely age and education attainment (human capital)

of an individual. On the other hand and related to the property itself, we will focus its current price, size and age.

The reference person represents the most financially knowledgeable member of a household and usually corresponds with the individual that earns the highest share of income. The HFCS determines who the reference person is depending on how the household is structured from a financial point of view. The age of the reference person is captured in the variable coded as "DHAGEH1" and the education of the reference person in the variable coded as "DHEDUH1". This variable represents the highest level of completed education of the reference person and is coded in the [1,8] interval, where "1" is associated with primary or no formal education and "8" is associated with a doctoral or parallel level.

The variable representing the current price of the main residence and coded "HB0700" was explained in detail in section 4.1 and will be used in this section as well. We have computed the age of the main residence as the difference between the year 2021 (year in which the HFCS wave was released) and the year of property acquisition (coded "HB0700"). We are aware of the fact that most surveys were most likely not conducted in the year 2021, but probably some time before, a circumstance that may not allow us to correctly represent the true age of the main residence at the time of the survey. Also, the year the property was acquired may not correspond to the year it was built. Nevertheless, we consider that the way our variable has been constructed is a reliable measure of the age of the property or at least a reliable measure of the time the property has been subject to capital depreciation since the household established their main residence in the property. We also consider that the time spanned between the year the survey was conducted and the year 2021 is not sufficient to alter any of the conclusions we may reach in our analysis. Lastly, the size of the household's main residence is captured in the variable coded as "HB0100".

With the results obtained at this point, we are able to give significant insights into the socioeconomic profile of the households that have experienced the most change in value in their main residences. However, given the rich source of data of the HFCS, we can deepen our analysis all the way down to the regional level and explore in which regions households have experienced the most changes in value. For that, we split the sample of each country according to the household's main residence location (captured in the variable "DHREGION") and compute the median absolute change in value per square meter of the property in order to control for its size. We also conduct an analog analysis comparing the change in value according to the age of the reference person.

4.2 Results

Tables number X and Y (see appendix) provide the main summary statistics for the net liquid assets and net worth of the 4915 Austrian and 121610 German surveyed households. Common

to both variables, we observe really high variability and dispersion in the data, reflected in variances and standard deviations of considerable dimensions. This indicates that the mean is not a good reflection of the sample's central tendency for any of the variables and that the median can serve our purposes better to that extent. With that taken into account, the median net liquid assets in Austria are of 25.000 Euros, lower than in Germany, in which they are of 55.475 Euros. This sharp contrast can also be observed if we take a look at the percentiles. The 10th, 25th and 50th percentiles are bigger in Germany than in Austria, which indicates that German households tend to own more net liquid assets at the bottom part of the distribution. The difference becomes bigger if we observe the 75th and 90th percentiles. In the case of Germany, higher levels of inequality at the top of the distribution can be observed. Regarding the net worth, the median net worth in Germany (549.500 Euros) is bigger than that of Austria (318.663,50 Euros). German households seem to have a higher net worth across percentiles, indicating higher levels of wealth. For both countries, the big differences observed between the 90th and 10th percentiles point towards a clear presence of inequality. Finally, figures X and Y (see appendix) depict a right-skewed distribution of both variables in both countries, which goes in line with the conclusions we have extracted from the analysis of the variable percentiles.

Figures W and Z show uni-variate regressions for Austria and Germany in which the dependent variable is the absolute change in real estate values and the independent variable are the household's net liquid assets. To correct for the presence of outliers, we have got rid of the top 1% of observations for each variable. Although the coefficients are positive and statistically significant (a one Euro increase in the net liquid assets leads to approximately 0,38 Euros of increase in main residence value in Austria and 0,20 Euros in Germany), the variability of the absolute change in real estate values can only marginally be explained by the net liquid assets alone, with R-squareds of 3,4% in Austria and 2,5% in Germany.

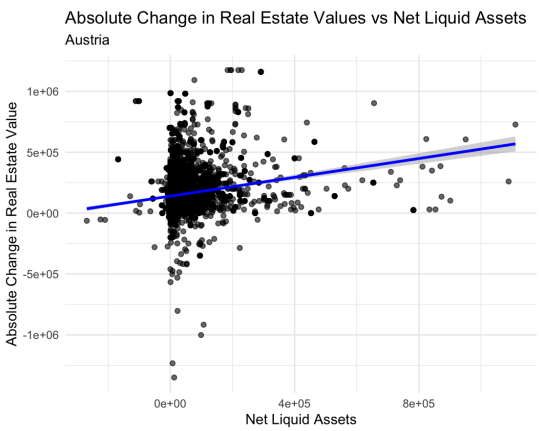

Figure 4
Data: HFCS 2021 wave. Own calculations.
R^2: 0,034. $\beta \approx 0,386$, $p < 0,01$, N = 4832.

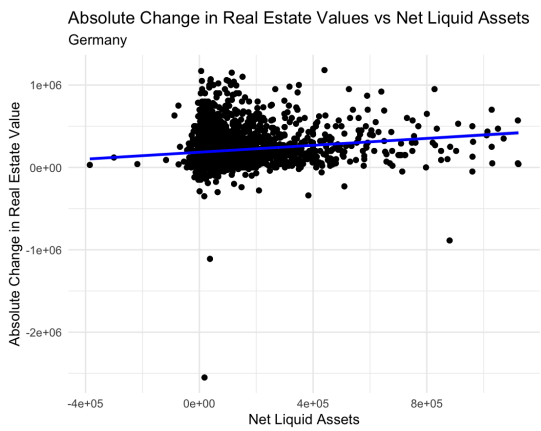

Figure 5
Data: HFCS 2021 wave. Own calculations.
R^2: 0,025. $\beta \approx 0,209$, $p < 0,01$, N = 11802.

Figures A and B on the other hand show uni-variate regressions for Austria and Germany in which the dependent variable is the absolute change in real estate values and the independent variable is the household's net worth. We've also got rid of the top 1% observations to correct for the presence of outliers. The coefficients are positive and statistically significant (a one Euro increase in net worth leads to approximately 0,19 Euros of increase in main residence value in Austria and 0,13 Euros in Germany). Even if the effect is probably small, around 22,1% of the variability of the absolute change in real estate values can be explained by the household's net worth in Austria and around 20,1% in Germany, which is about seven times as much as with the case of the net liquid assets.

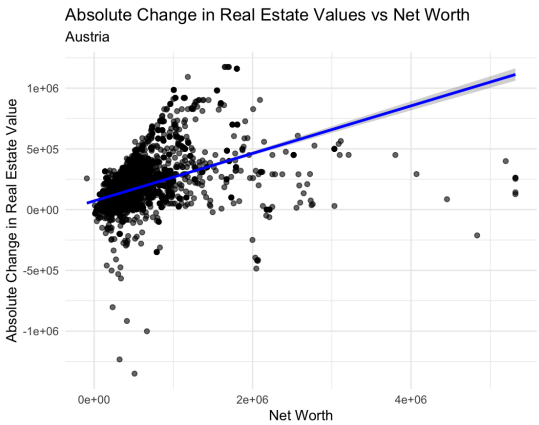

Figure 6
Data: HFCS 2021 wave. Own calculations.
R^2: 0,221. $\beta \approx 0,195$, $p < 0,01$, N = 4832.

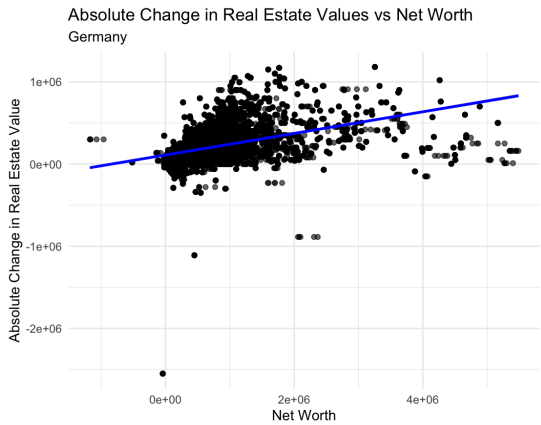

Figure 7
Data: HFCS 2021 wave. Own calculations.
R^2: 0,201. $\beta \approx 0,131$, $p < 0,01$, N = 11802.

Table X shows variable correlations with the absolute change in value of a household's main residence. The net liquid assets of a household seem to be weakly positively correlated with a change in value of their main residence (0,10 in AUT and 0,14 in GER). The correlation is much stronger if we consider the net worth (0,41 in AUT and 0,42 in DE). This suggests that households with a higher overall wealth tend to experience bigger increases in value in their main residences. However, the fact that these correlations are closer to zero than to one reveals that other variables must be determining the change in value. The age (0,21 in AUT and 0,19 in GER) and size (0,16 in AUT and 0,23 in GER) of the main residence are positively correlated with the property's absolute change in value. This result suggests the considered real estate properties tend to experience larger increases in value the older and bigger in size they are, although the correlation with the size of the property is not as strong for Austria as it is for Germany. This result complements the findings we obtained in section four, in which we saw that properties that were acquired earlier in time had experienced on average higher changes in value. Regarding the characteristics of the reference person, age seems to be only weakly positively correlated with the absolute change in value (0,20 for AUT and 0,13 for

GER), suggesting that the people who have owned their properties for a longer period of time are also the ones who have been exposed to suffering higher changes in value in their properties. Education attainment and gender of the reference person seem to not be correlated at all.

Overall, the correlations seem to be weak for all the considered variables except perhaps that of the net worth. Since the main residence is one of the most prominent non-financial assets in terms of value of a household, this is a result that shouldn't come as a surprise.

Table 3: Correlations with the Absolute Change in Value of a Household's Main Residence.

	Country	
Variable	**Austria**	**Germany**
Net liquid assets	0,10***	0,14***
Net worth	0,41***	0,42***
Age of the main residence	0,21***	0,19***
Size of the main residence	0,16***	0,23***
Age of the reference person	0,20***	0,13***
Education of the reference person	NS	0,09***
Gender of the reference person	NS	-0,03***

Data: HFCS 2021 wave. Own calculations. *** p<0,01, ** p<0,05, * p<0,1.
NS: Statistically not significant.

Tables X and Y collect variables that are correlated respectively with the household's net liquid assets and net worth. The age and education attainment of the reference person seem to be either very weakly positively correlated or not correlated at all with both variables in both countries suggesting that, for our specific sample, age and education doesn't come together with having a higher net worth or holding higher net liquid assets. Nevertheless, the current price of the main residence seems to be positively correlated with the household's net liquid assets (0,16 for AUT and 0,21 for GER) and very strongly positively correlated with the household's net worth (0,48 for AUT and 0,60 for GER). This result goes in line with the economic intuition we gave on section 5.1, indicating that households with more "cash-in-hand" and/or net worth tend to own more valuable properties.

Table 4: Variables correlated with a Household's Net Liquid Assets.

	Country	
Variable	**Austria**	**Germany**
Current price of the main residence	0,16***	0,21***
Age of the reference person	0,02*	0,08***
Education of the reference person	0,09***	0,11***

Data: HFCS 2021 wave. Own calculations. *** p<0,01, ** p<0,05, * p<0,1.

Table 5: Variables correlated with a Household's Net Worth.

	Country	
Variable	**Austria**	**Germany**
Current price of the main residence	0,48***	0,60***
Age of the reference person	NS	0,04***
Education of the reference person	0,03***	0,12***

Data: HFCS 2021 wave. Own calculations. *** p<0,01, ** p<0,05, * p<0,1.
NS: Statistically not significant.

Next, we have split our sample in age brackets of homogeneous size according to the age of the reference person (which in Austria and Germany takes values in the interval $[18, 85]$). Then we have computed the median absolute change per square meter of property that each age group has experienced in order to control for the size of the property. We have used box plots to depict the median experienced change per age category in each country. The result can be seen in figures X and Y. In the case of Austria, we observe how the inter-quartile ranges (represented by the size of the rectangles) is relatively homogeneous, with the age group in the bracket $[56, 65]$ showing perhaps slightly more variability. In the case of Germany, the inter-quartile ranges are not as homogeneous, with the age group of $[18, 25]$ exhibiting a really high variability. The presence of outliers is more not remarkable in Austria than in Germany, thus the different scales in the y-axis. Although the trends are different in both countries, we can observe how the median absolute change per square meter of main residence tends to increase with the age of the reference person. This is a result that goes in line with the correlations we have seen at the beginning of this section.

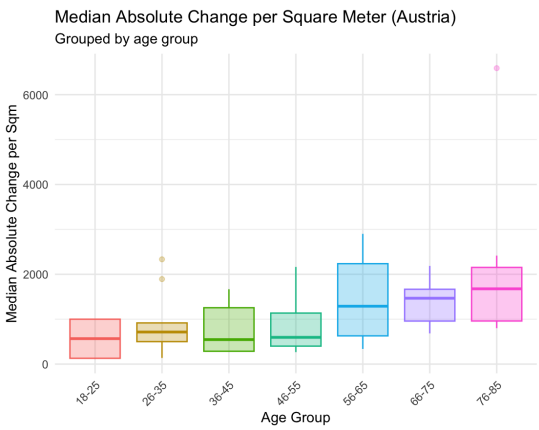

Figure 8
Data: HFCS 2021 wave. Own calculations.

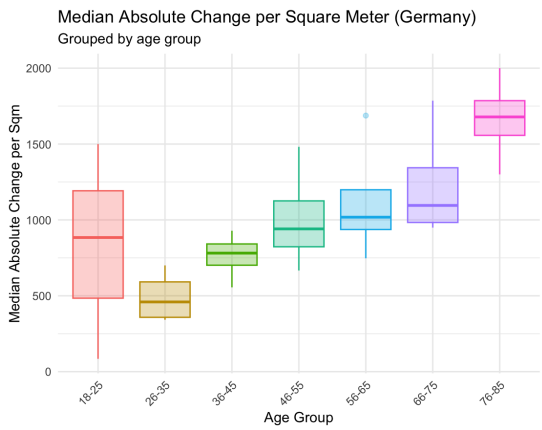

Figure 9
Data: HFCS 2021 wave. Own calculations.

We have also computed the median absolute change per square meter of main residence filtering by the region in which households reside. The graphical depiction of the changes can be seen in figures 7 and 8. In Austria, households living in the Federal States of Tyrol (2.222,22 Euros),

Vienna (1824,81 Euros) and Salzburg (1.645,72 Euros) are the ones who have experienced a higher median absolute change in value of their main residences. In Germany, it's been the households living in the Southern-Germany (1.583,33 Euros) who have experienced higher median changes in value. While the median changes in value per square meter experienced in the regions of Vienna and Salzburg are comparable in size with those occurred in Southern Germany, the changes in value that have occurred in Tyrol seem to be particularly high and could be driven in part by the presence of outliers. However, this are still findings which go in line with the consulted literature. Table X can be consulted in the appendix for a more in detail account of the results on this matter.

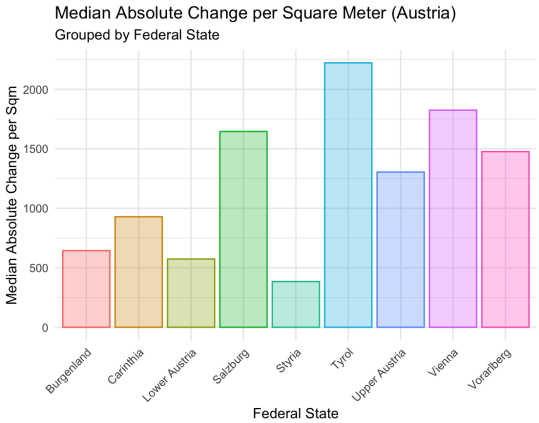

Figure 10
Data: HFCS 2021 wave. Own calculations.

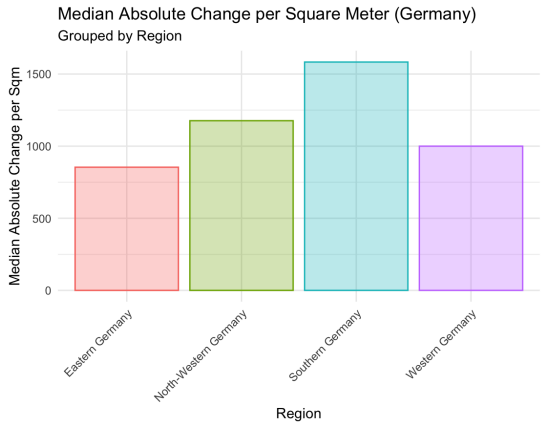

Figure 11
Data: HFCS 2021 wave. Own calculations.

After all the exposed results, we are able to identify the household profile who has suffered the most changes in value in their main residences for each country. In Austria this household corresponds with a household residing in the federal state of Tyrol, with a median main residence size of 175 m^2 that has suffered a median absolute change in main residence value of 856.759,50 Euros. Their median net worth is of 1.245.000 Euros and their median net liquid assets are of 28.000 Euros, with a median highest completed education level corresponding to upper secondary. In Germany this household corresponds to with a household residing in Southern-Germany with a median main residence size of around 140 m^2 that has suffered a median absolute change in main residence value of 250.000 Euros. This profile coincides with households in which the reference person is between 66-75 and 76-85 years old. Respectively, their median net worth are of 667.000 Euros and 727.000 Euros and their median net liquid assets are of 90.800 Euros and 82.000 Euros. Their median highest completed education corresponds to short-cycle tertiary education and upper secondary. The profile of households differs in both countries in the size of experienced change in value and in the amount of net liquid wealth and net worth that they hold.

5. Consumption and Savings Behavior of the Households

After conducting an analysis of households that experienced substantial changes in the value of their primary residence, we now turn to the final examination of their consumption and savings behavior in response to these observed fluctuations.

This analysis offers valuable insights into how a significant increase in real estate value influences household financial decisions, shaping their expenditure patterns and their propensity to allocate resources between consumption and savings. A fundamental question in macroeconomic research is whether an increase in real estate wealth leads to adjustments in household financial decisions, particularly in terms of expenditure and savings. The wealth effect hypothesis suggests that as household wealth increases, individuals may increase their consumption, assuming a positive marginal propensity to consume out of wealth. Conversely, some households may view the increase in real estate values as an opportunity to accumulate further savings.

5.1 Variables and methodology

Following the established variables in our previous subsections, we further create a new variable in order to represent our total consumption (total c) , representing overall household expenditures. We then further analyze total consumption divided by the number of household members, allowing for better comparability across households of different sizes. Similarly, we utilize a variable that represents their total net liquid assets as a mean to address their savings (totalnetliquidassets).

To investigate these behavioral adjustments, we utilize variables from the HFCS dataset, specifically focusing on household consumption and savings indicators (DOCOGOOD; DOFOODC; HI0210; DA2101). The primary variable representing total household consumption is the sum of all major expenditure categories, adjusted for household size to allow for cross-household comparability. This adjusted measure of per capita consumption provides a more standardized metric for analysis. Additionally, we examine household savings rates, defined as the difference between disposable income and total consumption.

To establish potential relationships, we analyze the correlation between real estate value changes and both consumption and savings behavior. Specifically, we categorize households based on the extent of their real estate appreciation and compare their consumption-to-income and savings-to-income ratios. Furthermore, we conduct regression analyses where the dependent variables are consumption and savings, while the key independent variable is the change in real estate value.

5.2 Results

The findings indicate no strong evidence of a systematic adjustment in household consumption or savings behavior in response to changes in real estate wealth. Figures 12 to 16 illustrate the distribution of consumption and savings levels across households that experienced substantial increases in property values. As seen in Table 12, the mean and median consumption levels remain relatively stable across households, regardless of whether their real estate values increased significantly. This suggests that, on average, households did not substantially alter their spending habits despite experiencing a rise in property wealth.

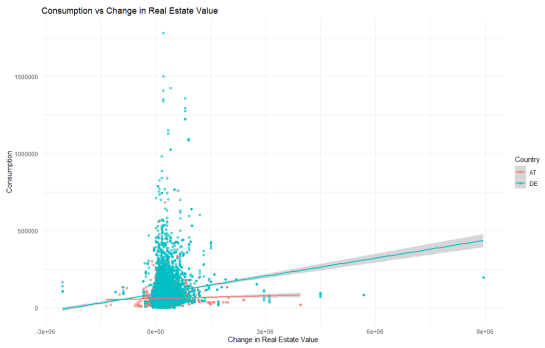

Figure 12
Data: HFCS 2021 wave. Own calculations.

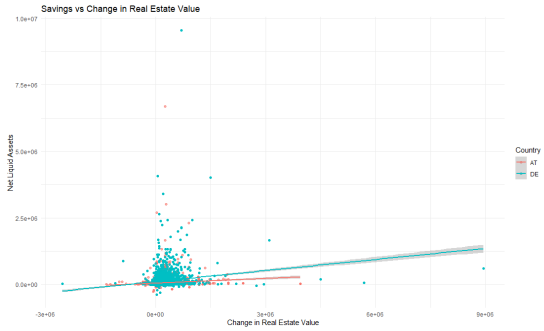

Figure 13
Data: HFCS 2021 wave. Own calculations.

Further examination of the savings rate, as depicted in Table 13, shows a similar trend. Households that benefited from large property value increases did not exhibit a significant increase in savings accumulation compared to those with modest or no property value appreciation. The correlation coefficients between real estate value changes and savings rates, presented in Table 15, reveal weak and statistically insignificant relationships, reinforcing the observation that changes in housing wealth have had limited influence on household financial behavior.

Additionally, Table 16 provides an analysis of household characteristics, including income levels, net liquid assets, and debt burdens, in relation to consumption and savings behavior. The results indicate that households with higher income and net liquid assets exhibit a more stable consumption pattern, which may explain the muted response to real estate appreciation.

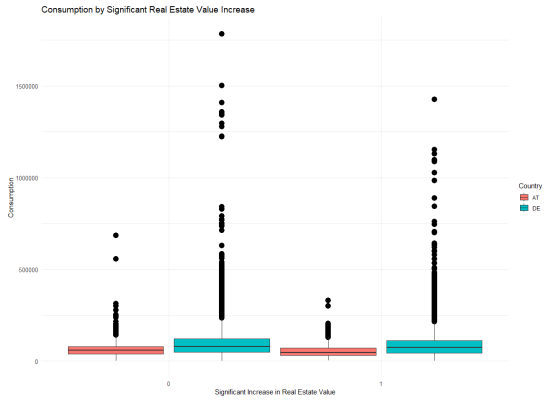

Figure 14
Data: HFCS 2021 wave. Own calculations.

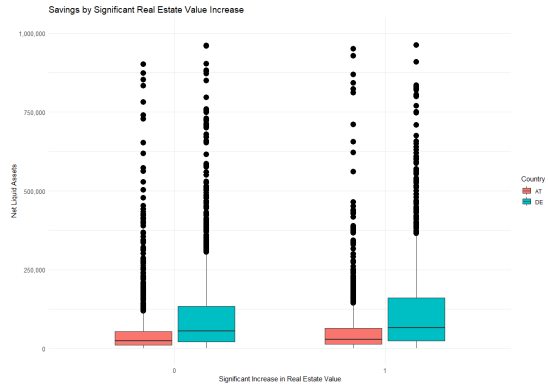

Figure 15
Data: HFCS 2021 wave. Own calculations.

While the results suggest a lack of significant behavioral adjustments, alternative explanations must be considered. One possibility is that households perceive real estate wealth as illiquid, making them less likely to adjust their consumption based on fluctuations in property values. Moreover, the age distribution of households may play a role; older homeowners who have accumulated substantial equity may prioritize wealth preservation over increased consumption.

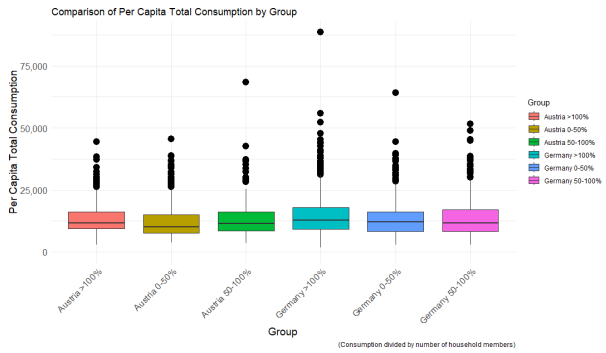

Figure 16
Data: HFCS 2021 wave. Own calculations.

To further explore the impact of real estate value changes on household expenditure, we introduce the average total consumption per capita across different household categories based on the extent of their real estate value changes. This provides a comparative perspective, highlighting that while some households with substantial real estate appreciation exhibit slightly higher per capita consumption levels, the overall differences remain statistically insignificant, reinforcing the earlier findings. For future research implementations, an instrumental variable regression could help identify causal effects and disentangle potential confounding factors.

6. Discussion

So far, our analysis has allowed us to gain a deeper understanding into house price developments in Austria and Germany through the lens of the latest available HFCS wave. Across all

stages of our analysis, we've observed how our results have been influenced by a high degree of data variability and dispersion, a trait reflected in high variances and standard deviations and a significant presence of outliers. However, although this needs to be taken into account when interpreting our results, we don't think this circumstance is sufficient to invalidate our findings, and can be mainly attributed to the way the survey data has been set up and compiled in terms of sample design and overall methodology.

In section 3, we have observed how a majority of households in Austria and Germany have suffered an increase in value in their main residences, with a significant share of them experiencing increases of over 50% and 100%. Furthermore, we've also seen how properties tend to have experienced more increase in value the earlier in time they were acquired, a finding that is backed by the computed correlations in section 4. These results go in line in qualitative terms with the ones obtained by Albacete et al. (2016), Kindermann et al. (2021), Schneider and Wagner (2015) and Schneider (2013), all of whom observed increases in real estate values in Austria and Germany, although different properties and time frames were considered in those studies.

Some aspects deserve attention from our side. First, and as pointed out by Albacete et al. (2016), we've computed price increases using non-transactional, elicited survey values. We also haven't been able to control for inflation and use real prices as Schneider and Wagner (2015) did, who obtained more modest price increases of 1,3% per year for Germany in the period 1980-2013. Moreover, there's nothing we've done in our analysis that could allow us to say whether these house price increases are justified by their fundamentals or are affected by some underlying speculative motives. Some of these considerations, such as having to work with "fictional" house prices cannot be avoided if one intends to work with this type of data. Regarding the others, further research and a more elaborate would be necessary to incorporate them into our analysis.

In section 4 we have observed how, for a household, owning more net worth and older, bigger main residences in which the reference person is older is correlated with having experienced higher changes in value in their main residences. We've also seen how owning more net worth is very positively correlated with owning a residence with a higher current price. The correlation is not so strong if we consider the net liquid assets. These findings are to be expected if we consider the life-cycle asset accumulation profile proposed by Kaplan and Violante (2014), in which the illiquid wealth present in the balance sheet of the household tends to grow with the age of the individuals. Additionally, we found that the regions of Southern Germany and Tyrol, Vienna and Salzburg in Austria have been the regions in which more property price increases have taken place. These results go in line with the ones obtained by Kindermann et al. (2021) and Schneider (2013).

It must be noted that the decision we made by opting for a mainly descriptive analysis in which no multivariate regression model determining the change in real estate values was specified, doesn't allow us to claim any causality. An insightful extension to this section should intend to specify such a model in order to give robustness to these findings.

Lastly, in section 5 we couldn't find any evidence suggesting that households that experienced an increase in their real estate wealth have adjusted their consumption and/or savings behavior. This is a result that comes in contrast to the findings obtained by Kindermann et al. (2021), in which they were able to observe an adjustment in savings and consumption behavior to increases in value of main residences. We believe that a extension to our analysis filtering by age of the reference person could reveal different consumption and savings patterns, as the life-cycle asset accumulation model of Kaplan and Violante (2014) suggests could be the case. Also, it must be noted that the size of our sample could have played a role in allowing us to find concise evidence pointing to changes in behavior in any direction.

7. Conclusion

This paper has studied house price developments in Austria and Germany with the help of the latest available HFCS 2021 wave. Our findings suggest that only a small proportion of households in Austria and Germany have experienced a decline in property value and that a substantial proportion of households in both Austria (43.3%) and Germany (37.2%) experienced property value increases exceeding 100%. Also properties acquired before 2010 were more likely to experience extreme value increases (over 100%), while those acquired after 2010 saw more moderate growth. Additionally, we've found that owning more net worth and older, bigger main residences in which the reference person is older is correlated with having experienced higher changes in value in their main residences. We've also seen how owning more net worth is very positively correlated with owning a residence with a higher current price and that households locates in Southern Germany and Tyrol in Austria are the ones who have experienced higher changes in value. Lastly, we haven't found any evidence suggesting that the households in our sample have changed their consumption and/or savings behavior when faced with an increase in the prices of their main residence.

Further extensions to this study should try to incorporate real prices to account for inflation and, in a more broader perspective, try to understand the reasons leading up to these changes in prices, for which a multivariate econometric specification could be a valuable addition to the analysis. Lastly, exploring the role of age profiles in determining changes in household consumption and savings patterns when faced with an increase in value of their illiquid assets should give more precise insights into how has their behavior in this sense really changed.

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Appendix: Figures and Tables

Table 6: Net Liquid Assets (Euros) - Summary Statistics.

Variable	Country	
Net Liquid Assets	**Austria**	**Germany**
Mean	56160,15	139927,72
Median	25000	55475
Standard Deviation	150016,40	330780,30
Variance	22,504e+09	109,415e+09
10th Percentile	2217,60	3000
25th Percentile	10625,50	18000
50th Percentile	25000	55475
75th Percentile	58700	145000
90th Percentile	126200	328000

Data: HFCS 2021 wave. Own calculations.
Number of observations: 4915 Austria, 12160 Germany.

Table 7: Net Worth (Euros) - Summary Statistics.

Variable	Country	
Net Worth	**Austria**	**Germany**
Mean	507283,80	885015,30
Median	318663,50	549500
Standard Deviation	755737,90	1262783,20
Variance	5,711e+11	1,594e+12
10th Percentile	156931,60	175000
25th Percentile	213870,50	324000
50th Percentile	318663,50	549500
75th Percentile	535448,50	972500
90th Percentile	940875,60	1739300

Data: HFCS 2021 wave. Own calculations.
Number of observations: 4915 Austria, 12160 Germany.

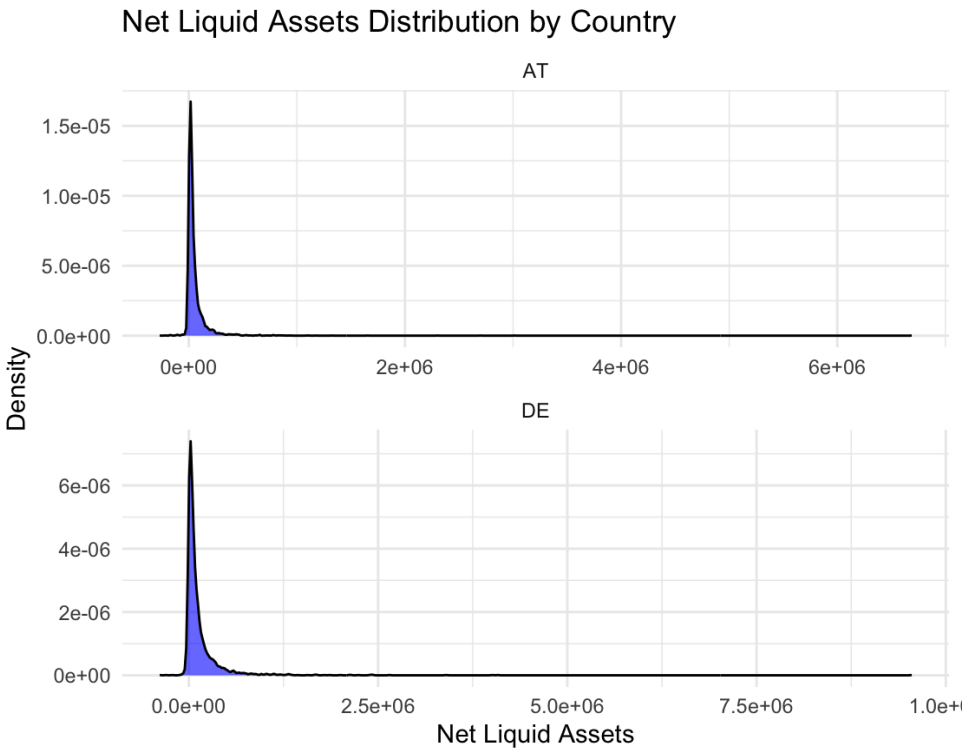

Figure 17
Data: HFCS 2021 wave. Own calculations.

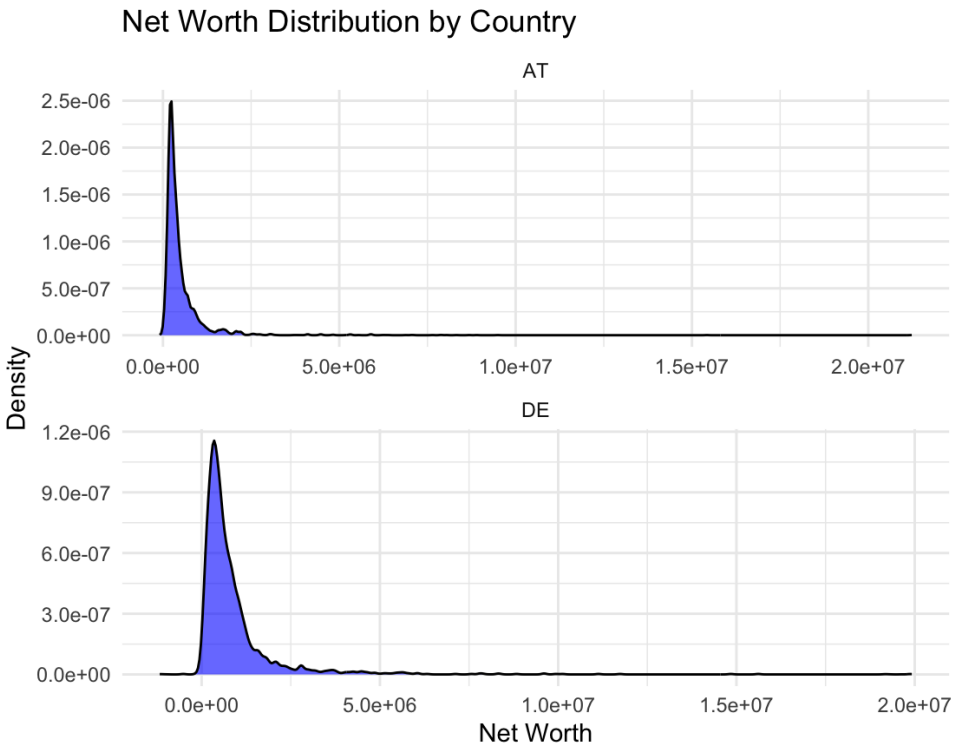

Figure 18
Data: HFCS 2021 wave. Own calculations.

Table 8: Median Absolute Change in Value per Square meter (Euros) and Median Main Residence Size (Sqm) in Austria and Germany (Grouped by Regions and Federal States)

Region	**Median Absolute Change per Sqm (Euros)**	**Median Main Residence Size (Sqm)**
Eastern Germany	854,16	129,45
North-Western Germany	1176,47	145,24
Southern Germany	1583,33	146
Western Germany	1000	147,36
Burgenland	644,02	142,50
Carinthia	928,57	136,08
Lower Austria	573,80	142,51
Salzburg	1645,72	118,51
Styria	384,61	132,39
Tyrol	2222,22	122,85
Upper Austria	1304,30	146,53
Vienna	1824,81	113,02
Vorarlberg	1476,18	137,39

Data: HFCS 2021 wave. Own calculations. Eastern Germany: Mecklenburg-West Pomerania, Saxony Anhalt, Brandenburg, Berlin, Thuringia, Saxony; North-Western Germany: Lower Saxony, Schleswig-Holstein, Hamburg, Bremen; Southern Germany: Bavaria, Baden-Württemberg, Hesse; Western Germany: North Rhine-Westphalia, Rhineland-Palatinate, Saarland.

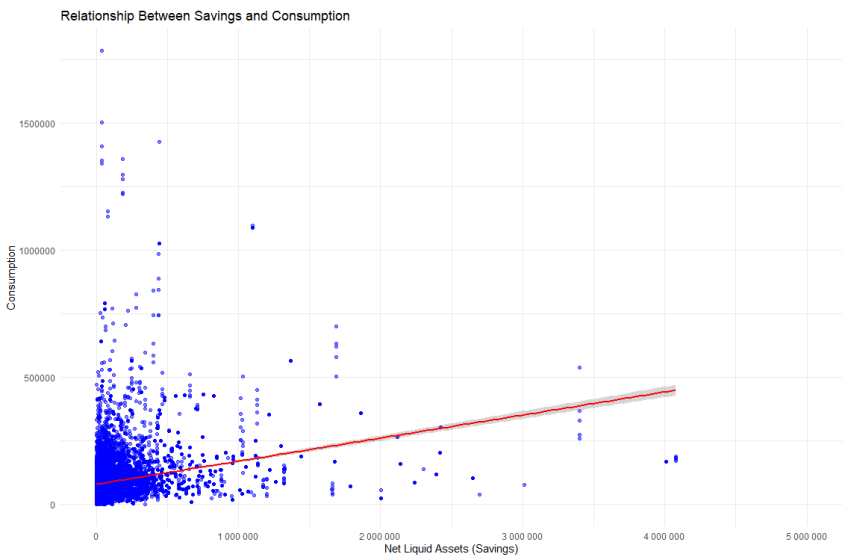

Figure 19
Data: HFCS 2021 wave. Relationship between Savings and Consumption for HH with significant increase in main residence value. Own calculations.